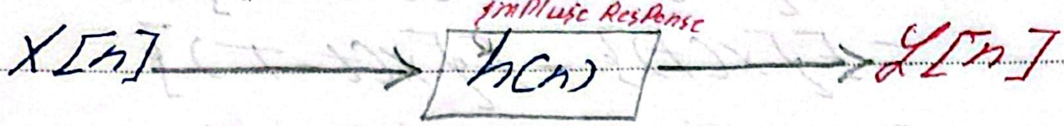
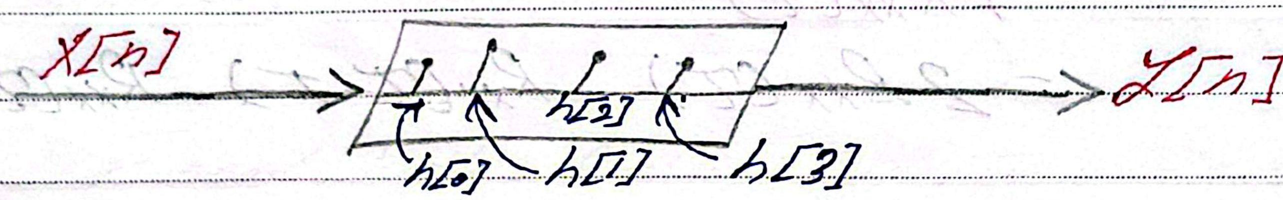
✱ Convolution :-



$$y[n] = x[n] \circledast h[n]$$
$$= \sum_{k=0}^{\infty} x[n-k]\, h[k]$$

$$y[n] = x[n]h[0] + x[n-1]h[1] + x[n-2]h[2] + x[n-3]h[3]$$



Ex. $x[n] = \{4, 1, 2, 5\}$ $\quad h[n] = \{1, 2, -1\}$

x	4	1	2	5		
h		1	2	-1		
	4	1	2	5		
		8	2	4	10	
			-4	-1	-2	-5
	4	9	0	8	8	-5

الموضوع: .... التاريخ: / /

Cross Correlation

Ex: $h[n] = [1, 2, 3]$ $x[n] = [4, 5]$

step 1

	1	2	3
	4	5	0
	4	10	0 = 14

step 2

	1	2	3
x[n-1]	0	4	5
	0	8	15 = 23

step 3

	1	2	3	0
x[n-2]	0	0	4	5
	0	0	12	0 = 12

step 4

	0	1	2	3
$x[n+\boxed{1}]$	4	5	0	0
	0	5	0	0 = 5

step 5

	0	0	1	2	3
x[n+2]	4	5	0	0	0
	0	0	0	0	0 = 0

h(n) x(n) h(n) x(n-2)

$Corr(h, x) = [0 \quad 5 \quad 14 \quad 23 \quad 12]$

h(n) x(n+2) h(n) x(n+1) h(n) x(n-1)

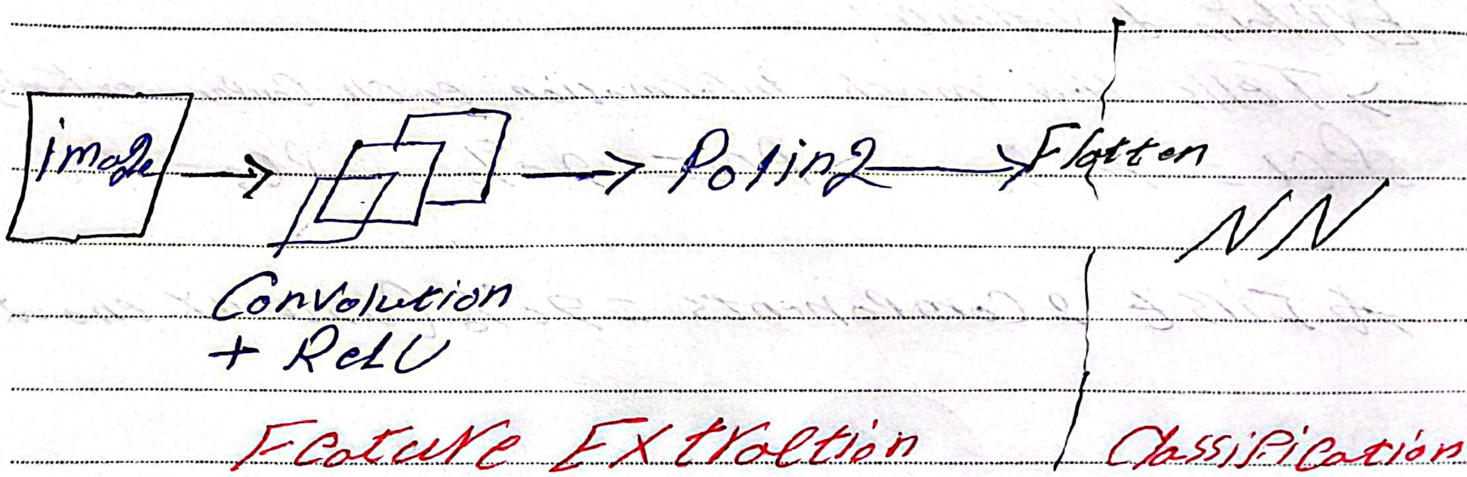


image
Convolution
+ ReLU
Poling
Flatten
NN
Feature Extraction
Classification